

MT100 and MT200 protocol

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i. Communication Protocol

Introduction

This document defines instructions about interface protocol on application layer of vehicles GPS tracker and location-based service platform. Related interface protocol only applies in the interaction between the platform and the position terminal.

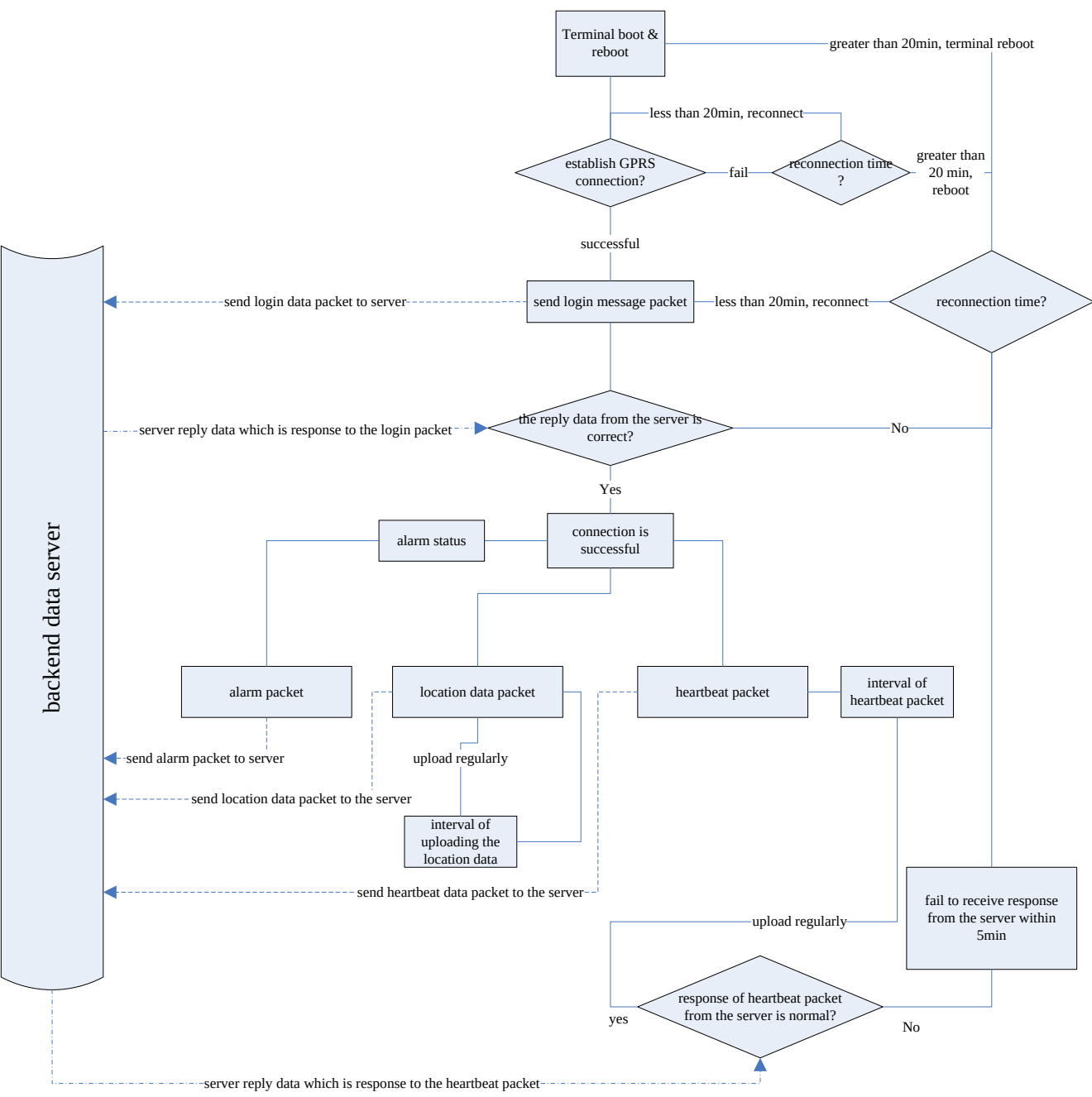
ii. Terms, Definitions

Terms, Abbreviation	Definition in English	Definition in Chinese
CMPP	China Mobile Peer to Peer	中国移动点对点协议
GPS	Global Positioning System	全球卫星定位系统
GSM	Global System for Mobile Communication	全球移动通信系统
GPRS	General Packet Radio Service	通用无线分组业务
TCP	Transport Control Protocol	传输控制协议
LBS	Location Based Services	辅助定位服务
IMEI	International Mobile Equipment Identity	国际移动设备识别码
MCC	Mobile Country Code	移动用户所属国家代号
MNC	Mobile Network Code	移动网号码
LAC	Location Area Code	位置区码
Cell ID	Cell Tower ID	移动基站
UDP	User Datagram Protocol	用户数据报协议
SOS	Save Our Ship/Save Our Souls	遇难求救信号
CRC	Cyclic Redundancy Check	循环冗余校验
NITZ	Network Identity and Time Zone,	时区
GIS	Geographic Information System	地理信息系统

iii. Basic Rules

1. If a GPRS connection is established successfully, the terminal will send a first login message packet to the server and, within five seconds, if the terminal receives a data packet responded by the server, the connection is considered to be a normal connection. The terminal will begin to send location information (i.e., GPS, LBS information package). A status information package will be sent by the terminal after three minutes to regularly confirm the connection.
2. If the GPRS connection is established unsuccessfully, the terminal will not be able to send the login message packet. The terminal will start schedule reboot in twenty minutes if the GPRS connection is failed three times. Within twenty minutes, if the terminal successfully connects to the server and receives the data packet from the server as the server's response to the login message packet sent by the terminal, the schedule reboot will be off and the terminal will not be rebooted; otherwise, the terminal will be rebooted automatically in twenty minutes.
3. After receiving the login message packet, the server will return a response data packet. If the terminal doesn't receive packet from the server within five seconds after sending the login message packet or the status information package, the current connection is regarded as an abnormal connection. The terminal will start a retransmission function for GPS tracking data, which will cause the terminal to disconnect the current GPRS connection, rebuild a new GPRS connection and send a login message packet again.
4. If the connection is regarded to be abnormal, and the data packet as a response from the server is failed to be received three times after a connection is established and a login message packet or status information package is sent, the terminal will start schedule reboot and the scheduled time is ten minutes. Within ten minutes, if the terminal successfully connects to the server and receives the data packet responded by the server, the schedule reboot will be off and the terminal will not be rebooted; otherwise, the terminal will be rebooted automatically in ten minutes.
5. In case of the normal connection, the terminal will send a combined information package of GPS and LBS to the server after the GPS information is changed; and the server may set a default protocol for transmission by using commands.
6. To ensure the effectiveness of the connection, the terminal will send status information to the server at regular intervals, and the server will return response data packets to confirm the connection.
7. For the terminal which doesn't register an IMEI number, the server will reply the terminal with a login request response and heartbeat packet response, rather than directly disconnect the connection. (If the connection is directly disconnected or the server doesn't reply to the terminal, it will lead to a continuous reconnected by the terminal and the GPRS traffic will be consumed heavily.

Data Flow Diagram



iv. Data Packet Format

The communication is transferred asynchronously in bytes.

The total length of packets is (10+N) Bytes.

Format	Length(Byte)
Start Bit	2
Packet Length	1
Protocol Number	1
Information Content	N
Information Serial Number	2
Error Check	2
Stop Bit	2

4.1. Start Bit

Fixed value in HEX 0x78 0x78.

4.2. Packet Length

Length = Protocol Number + Information Content + Information Serial Number + Error Check, totally (5+N)Bytes, because the Information Content is a variable length field.

4.3. Protocol Number

Type	Value
Login Message	0x01
Location Data	0x22
Heartbeat packet	0x13
Calibration time Packet	0x8A
Alarm data	0x26
Cell information packet	0x28
General Information Package	0x94
Command information sent by the server to the terminal	0x80
Server reply	0x15

4.4. Information Contents

The specific contents are determined by the protocol numbers corresponding to different applications.

4.5. Information Serial Number

The serial number of the first GPRS data (including status packet and data packet such as GPS, LBS) sent after booting is '1', and the serial number of data sent later at each time will be automatically added '1'.

4.6. Error Check

A check code may be used by the terminal or the server to distinguish whether the received information is error or not. To prevent errors occur during data transmission, error check is added to against data misoperation, so as to increase the security and efficiency of the system. The check code is generated by the CRC-ITU checking method.

The check codes of data in the structure of the protocol, from the Packet Length to the Information Serial Number (including "Packet Length" and "Information Serial Number") , are values of CRC-ITU.

CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet.

4.7. Stop Bit

Fixed value in HEX 0x0D 0x0A.

v. Details about Data Packet sent by Server to Terminal

The commonly used information packages sent by the terminal and those sent by the server will be interpreted separately.

5.1. Login Message Packet

5.1.1. Terminal Sending Data Packet to Server

The login message packet is used to be sent to the server with the terminal ID so as to confirm the established connection is normal or not.

	Description	Bits	Example
Login Message Packet(18 Byte)	Start Bit	2	<u>0x78 0x78</u>
	Packet Length	1	<u>0x11</u>
	Protocol Number	1	<u>0x01</u>
	Terminal ID	8	<u>0x01 0x23 0x45 0x67 0x89 0x01 0x23 0x45</u>
	Device type	2	<u>0x20,0x03</u>
	Time zone and language	2	<u>Refer to the table below</u>
	Information Serial Number	2	<u>0x00 0x01</u>
	Error Check	2	<u>0x8C 0xDD</u>
	Stop Bit	2	<u>0x0D 0x0</u>

Time zone and language

bit15	Time zone Convert example GMT+8:25 8*100+25=825=0x0339 GMT-12:45 12*100+45=1245=0x04DD	
bit14		
bit13		
bit12		
bit11		
bit10		
bit9		
bit8		
bit7		
bit6		
bit5		
bit4		
bit3	0 East Time Zone	1 West Time Zone
bit2	reverse	
bit1	Language bit	1
bit0	Language bit	0

5.1.1.1. Start Bit

For details see Data Packet Format section 4.1.

5.1.1.2. Packet Length

For details see Data Packet Format section 4.2.

5.1.1.3. Protocol Number

For details see Data Packet Format section 4.3.

5.1.1.4. Terminal ID

The terminal ID applies IMEI number of 15 bits.

Example: if the IMEI is 123456789012345,
the terminal ID is 0x01 0x23 0x45 0x67 0x89 0x01 0x23 0x45.

5.1.1.5. Information Serial Number

For details see Data Packet Format section 4.5.

5.1.1.6. Error Check

For details see Data Packet Format section 4.6.

5.1.1.7. Stop Bit

For details see Data Packet Format section 4.7.

5.1.2. Server Responds the Data Packet

	Description	Bits	Example
Login Message Packet (18 Byte)	Start Bit	2	<u>0x78 0x78</u>
	Packet Length	1	<u>0x05</u>
	Protocol Number	1	<u>0x01</u>
	Information Serial Number	2	<u>0x00 0x01</u>
	Error Check	2	<u>0xD9 0xDC</u>
	Stop Bit	2	<u>0x0D 0x0A</u>

The response packet from the server to the terminal: the protocol number in the response packet is identical to the protocol number in the data packet sent by the terminal.

5.1.2.1. Start Bit

For details see Data Packet Format section 4.1.

5.1.2.2. Packet Length

For details see Data Packet Format section 4.2.

5.1.2.3. Protocol Number

For details see Data Packet Format section 4.3.

5.1.2.4. Information Serial Number

For details see Data Packet Format section 4.5.

5.1.2.5. Error Check

For details see Data Packet Format section 4.6.

5.1.2.6. Stop Bit

For details see Data Packet Format section 4.7.

5.1.3. Examples

Examples of the login message packet sent by the terminal to the server and the response packet sent by the server to the terminal are as follows: (in the examples the terminal ID is 752533678900242

login package 78 78 11 01 07 52 53 36 78 90 02 42 70 00 32 01 00 05 12 79 0D 0A

response package 78 78 05 01 00 05 9F F8 0D 0A

5.2. Location Data Packet (combined information package of GPS and LBS)

5.2.1. Terminal Sending Location Data Packet to Server

Format		Length(Byte)	Example
Information Content	Start Bit	2	0x78 0x78
	Packet Length	1	0x26
	Protocol Number	1	GT220:0x22 GT06N:0x12
	Date Time	6	0x0B 0x08 0x1D 0x11 0x2E 0x10
	Quantity of GPS information satellites	1	0xCF
	Latitude	4	0x02 0x7A 0xC7 0xEB
	Longitude	4	0x0C 0x46 0x58 0x49
	Speed	1	0x00
	Course, Status	2	0x14 0x8F
	MCC	2	0x01 0xCC
	MNC	1	0x00
	LAC	2	0x28 0x7D
	Cell ID	3	0x00 0x1F 0xB8
	ACC	1	bit0=1:ACC on bit0=0:ACC off bit1=1: Air-conditioner on bit1=0: Air-conditioner off
	Data type	1	0x00 Upload by time interval 0x01 Upload by distance interval 0x02 inflection point upload 0x03 ACC status change upload 0x04 change from motion to stationary state, 0xF0 Air-conditioner status change upload
	GPS real time or Supplementary transmission	1	0x00:real time 0x01: Supplementary transmission
	mileage	4	Unit: meter
	Serial Number	2	0x00 0x03
	Error Check	2	0x80 0x81
	Stop Bit	2	0x0D 0x0A

5.2.1.1. Start Bit

For details see Data Packet Format section 4.1.

5.2.1.2. Packet Length

For details see Data Packet Format section 4.2.

5.2.1.3. Protocol Number

For details see Data Packet Format section 4.3.

5.2.1.4. Date Time

Format	Length(Byte)	Example
Year	1	0x0A
Month	1	0x03

Day	1	0x17
Hour	1	0x0F
Minute	1	0x32
Second	1	0x17

Example: 2010-03-23 15:30:23

Calculated as follows: 10(Decimal)=0A(Hexadecimal)

3 (Decimal)=03(Hexadecimal)

23(Decimal)=17(Hexadecimal)

15(Decimal)=0F(Hexadecimal)

50(Decimal)=32(Hexadecimal)

23(Decimal)=17(Hexadecimal)

Then the value is: 0x0A 0x03 0x17 0x0F 0x32 0x17

5.2.1.5. Length of GPS information, quantity of positioning satellites

The field is 1 Byte displayed by two hex digits, wherein the first one is for the length of GPS information and the second one for the number of the satellites join in positioning.

Example: if the value is 0xCB, it means the length of GPS information is 12 and the number of the positioning satellites is 11.

(C = 12Bit Length , B = 11 satellites)

5.2.1.6. Latitude

Four bytes are consumed, defining the latitude value of location data. The range of the value is 0-162000000, indicating a range of 0 °-90 °. The conversion method thereof is as follow:

converting the value of latitude and longitude output by GPS module into a decimal based on minute; multiplying the converted decimal by 30000; and converting the multiplied result into hexadecimal.

Example: 22°32.7658'=(22X60+32.7658)X30000=40582974, then converted into a hexadecimal number

40582974(Decimal)= 26B3F3E(Hexadecimal)

at last the value is 0x02 0x6B 0x3F 0x3E.

5.2.1.7. Longitude

Four bytes are consumed, defining the longitude value of location data. The range of the value is 0-324000000, indicating a range of 0 °-180 °.

The conversion method herein is same to the method mentioned in Latitude (see section 5.2.1.6).

5.2.1.8. Speed

One byte is consumed, defining the running Speed of GPS. The value ranges from 0x00 to 0xFF indicating a range from 0 to 225km/h.

e.g. 0x00 represents 0 km/h.

0x10 represents 16km/h.

0xFF represents 255 km/h.

5.2.1.9. Course Status

Two bytes are consumed, defining the running direction of GPS. The value ranges from 0 ° to 360 ° measured clockwise from north of 0 °.

BYTE_1	Bit7	0
	Bit6	0
	Bit5	GPS real-time/differential positioning
	Bit4	GPS having been positioning or not
	Bit3	East Longitude, West Longitude
	Bit2	South Latitude, North Latitude
	Bit1	Course
	Bit0	
BYTE_2	Bit7	
	Bit6	
	Bit5	
	Bit4	
	Bit3	
	Bit2	
	Bit1	
	Bit0	

Note: The status information in the data packet is the status corresponding to the time bit recorded in the data packet.

For example: the value is 0x15 0x4C, the corresponding binary is 00010101 01001100,

BYTE 1 Bit7 0

BYTE 1 Bit6 0

BYTE_1 Bit5	0 (real time GPS)
-------------	-------------------

BYTE_1 Bit4	1 (GPS has been positioned)
-------------	-----------------------------

BYTE_1 Bit3 0 (East Longitude)

BYTE_1 Bit2	1 (North Latitude)
-------------	--------------------

BYTE	1	Bit1	0
------	---	------	---

BYTE_1 Bit0	1
-------------	---

BYTE 2 Bit7	0
-------------	---

BYTE 2 Bit7	1
-------------	---

BYTE 2 Bit7 0 → Course 332 °(0101001100 in Binary, or 332 in decimal)

BYTE 2 Bit7 0

BYTE_2 Bit7	1
-------------	---

BYTE 2 Bit7	1
-------------	---

BYTE 2 Bit7	0
-------------	---

BYTE 2 Bit7	0
-------------	---

which means GPS tracking is on, real time GPS, location at north latitude, east longitude and the course is 332 °.

5.2.1.10. MCC

The country code to which a mobile user belongs, i.e., Mobile Country Code(MCC).

Example: Chinese MCC is 460 in decimal, or 0x01 0xCC in Hex (that is, a decimal value of 460 converting into a hexadecimal value, and 0 is added at the left side because the converted hexadecimal value is less than four digits).

Herein the range is 0x0000 ~ 0x03E7.

5.2.1.11. MNC

Mobile Network Code(MNC)

Example: Chinese MNC is 0x00.

5.2.1.12. LAC

Location Area Code (LAC) included in LAI consists of two bytes and is encoded in hexadecimal. The available range is 0x0001-0xFFFE, and the code group 0x0000 and 0xFFFF cannot be used. (see GSM specification 03.03, 04.08 and 11.11).

5.2.1.13. Cell ID

Cell Tower ID (Cell ID), which value ranges from 0x000000 to 0xFFFFFF.

5.2.1.14. Mileage Statistics

Mileage statistics, statistics mileage calculation value, 16 binary displa.

5.2.1.15. Information Serial Number

For details see Data Packet Format section 4.5.

5.2.1.16. Error Check

For details see Data Packet Format section 4.6.

5.2.1.17. Stop Bit

For details see Data Packet Format section 4.7.

5.2.2. Examples of Packet Sent from Terminal to Server

7878 26 22 120913082434 CB 0436A7D0 0B7096D0 08 1413 01CC 00 9523 0060A5 01 00 00 0357DBB8 003E EB59 0D0A

5.3. Alarm Packet (GPS, LBS, combined status information packet)

5.3.1. Server Sending Alarm Data Packet to Server

Format		Length(Byte)	Example
Information Content	Start Bit	2	0x78 0x78
	Packet Length	1	0x25
	Protocol Number	1	GT220:0x26 GT06N:0x16
	GPS Information	Date Time	0x0B 0x08 0x1D 0x11 0x2E 0x10
		Quantity of GPS information satellites	0xCF
		Latitude	0x02 0x7A 0xC7 0xEB
		Longitude	0x0C 0x46 0x58 0x49
		Speed	0x00
		Course, Status	0x14 0x8F
	LBS Information	LBS Length	0x08
		MCC	0x01 0xCC
		MNC	0x00
		LAC	0x28 0x7D
		Cell ID	0x00 0x1F 0xB8
	status Information	Terminal Information Content	1
		Voltage Level	1
		GSM Signal Strength	1
		Alarm/Language	2
	Serial Number	2	0x00 0x03
	Error Check	2	0x80 0x81
	Stop Bit	2	0x0D 0x0A

Alarm packet is consisted by adding status information to location packet, so does the encoding format of the protocol.

5.3.1.1. Start Bit

For details see Data Packet Format section 4.1.

5.3.1.2. Packet Length

For details see Data Packet Format section 4.2.

5.3.1.3. Protocol Number

For details see Data Packet Format section 4.3.

5.3.1.4. Date Time

For details see Location Data Packet Format section 5.2.1.4.

5.3.1.5. Length of GPS information, quantity of positioning satellites

For details see Location Data Packet Format section 5.2.1.5.

5.3.1.6. Latitude

For details see Location Data Packet Format section 5.2.1.6.

5.3.1.7. Longitude

For details see Location Data Packet Format section 5.2.1.7.

5.3.1.8. Speed

For details see Location Data Packet Format section 5.2.1.8.

5.3.1.9. Status and Course

For details see Location Data Packet Format section 5.2.1.9.

5.3.1.10. MCC

For details see Location Data Packet Format section 5.2.1.10.

5.3.1.11. MNC

For details see Location Data Packet Format section 5.2.1.11.

5.3.1.12. LAC

For details see Location Data Packet Format section 5.2.1.12.

5.3.1.13. Cell ID

For details see Location Data Packet Format section 5.2.1.13.

5.3.1.14. Terminal Information

One byte is consumed, defining various status information of the mobile phone.

Bit		Code Meaning
BYTE	Bit7	1: oil and electricity disconnected
		0: gas oil and electricity connected
	Bit6	1: GPS tracking is on
		0: GPS tracking is off
	Bit3~ Bit5	100: SOS
		011: Low Battery Alarm
		010: Power Cut Alarm
		001: Shock Alarm
		000: Normal
	Bit2	1: Charge On
		0: Charge Off
	Bit1	1: ACC high
		0: ACC Low
	Bit0	1: Activated
		0: Deactivated

Example: 0x44, corresponding binary value is 01000100,

indicates that the status of the terminal is: oil and electricity connected, GPS tracking is on, normal without any alarm, charge on, ACC is low, and deactivated.

5.3.1.15. Voltage Level

The arrange is 0~6 defining the voltage is from low to high.

0: No Power (shutdown)

1: Extremely Low Battery (not enough for calling or sending text messages, etc.)

2: Very Low Battery (Low Battery Alarm)

3: Low Battery (can be used normally)

4: Medium

5: High

6: Very High

Example: 0x02 indicates very low battery and a Low Battery Alarm is sending.

5.3.1.16. GSM Signal Strength Levels

0x00: no signal;

0x01: extremely weak signal;

0x02: very weak signal;

0x03: good signal;

0x04: strong signal.

Example: 0x03 indicates the GSM signal is good.

5.3.1.17. Alarm/Language

0x00 (former bit) 0x01 (latter bit)

former bit: terminal alarm status (suitable for alarm packet and electronic fence project)

latter bit: the current language used in the terminal

former bit	0x00: normal
	0x01: SOS alarm
	0x02: Power Cut Alarm
	0x03: Vibration alarm
	0x06 Over speed alarm
	0x09 Moving alarm
	0x0E Low battery(external) alarm
	0x29 Accelerate Alarm
	0x2A Sharp right Turn Alarm
	0x2B Sharp left Turn Alarm
	0x2C Collision alarm
	0x30 Sharp Decelerate Alarm
	0xE0 turn over
latter bit	0x01: Chinese
	0x02: English

Examples:

No Alarm and Language is Chinese: 0x00 0x01

No Alarm and Language is English: 0x00 0x02

To increase the reliability of alarm information, labeling the alarm information repeatedly; in most cases, the alarm information keeps consistent with information of former terminal, while the inconsistencies are as follows:

- A. Low Battery Alarm occurred in the information of the terminal**
- B. Fence in and out Alarm in the Alarm/Language information**

5.3.1.18. Information Serial Number

For details see Data Packet Format section 4.5.

5.3.1.19. Error Check

For details see Data Packet Format section 4.6.

5.3.1.20. Stop Bit

For details see Data Packet Format section 4.7.

5.3.2. Examples

Examples of terminal transmission							
78 78 25 26 0B 0B 0F 0E 24 1D CF 02 7A C8 87 0C 46 57 E6 00 14 02 09 01 CC 00 28 7D 00 1F 72 65 06 04 01 01 00 36 56 A4 0D 0A							
Explain							
<u>0x78 0x78</u>	<u>0x25</u>	<u>0x26</u>	<u>0x0B 0x0B 0x0F 0x0E 0x24 x01D</u>		<u>0xCF</u>	<u>0x02 0x7A 0xC8 0x87</u>	
Start Bit	Length	Protocol No.	Date Time		Quantity of GPS information satellites	Latitude	
<u>0x0C 0x46 0x57 0xE6</u>	<u>0x00</u>	<u>0x14 0x02</u>	<u>0x09</u>	<u>0x01 0xCC</u>	<u>0x00</u>	<u>0x28 0x7D</u>	<u>0x00 0x1F 0x72</u>
Longitude	Speed	Course Status	LBS Length	MCC	MNC	LAC	Cell ID
<u>0x65</u>	<u>0x06</u>	<u>0x04</u>	<u>0x01 0x01</u>	<u>0x00 0x36</u>	<u>0x56 0xA4</u>	<u>0x0D 0x0A</u>	
Terminal Information Content	Voltage Level	GSM Signal Strength	Alarm/Language	Serial No.	Error Check	Stop Bit	

Note: The status information in the data packet is the status corresponding to the time bit recorded in the data packet.

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5.4. Heartbeat Packet (status information packet)

Heartbeat packet is a data packet to maintain the connection between the terminal and the server.

5.4.1. Terminal Sending Heartbeat Packet to Server

Format			Length (Byte)
Information Content	Start Bit		2
	Packet Length		1
	Protocol Number		1
	Status Information	Terminal Information Content	1
		Voltage Level	1
		GSM Signal Strength	1
		Alarm/Language	2
	Serial Number		2
	Error Check		2
	Stop Bit		2

5.4.1.1. Start Bit

For details see Data Packet Format section 4.1.

5.4.1.2. Packet Length

For details see Data Packet Format section 4.2.

5.4.1.3. Protocol Number

For details see Data Packet Format section 4.3.

5.4.1.4. Terminal Information

One byte is consumed defining for various status information of the mobile phone.

Bit		Code Meaning
BYTE	Bit7	1: oil and electricity disconnected
		0: gas oil and electricity
	Bit6	1: GPS tracking is on
		0: GPS tracking is off
	Bit3~ Bit5	100: SOS
		011: Low Battery Alarm
		010: Power Cut Alarm
		001: Shock Alarm
		000: Normal
	Bit2	1: Charge On
		0: Charge Off
	Bit1	1: ACC high
		0: ACC Low
	Bit0	1: Activated
		0: Deactivated

Example: 0x44, corresponding binary value is 01000100,

indicates that the status of the terminal is: oil and electricity connected, GPS tracking is on, normal without any alarm, charge on, ACC is low, and deactivated.

5.4.1.5. Voltage Level

The arrange is 0~6 defining the voltage is from low to high.

0: No Power (shutdown)

- 1: Extremely Low Battery (not enough for calling or sending text messages, etc.)
- 2: Very Low Battery (Low Battery Alarm)
- 3: Low Battery (can be used normally)
- 4: Medium
- 5: High
- 6: Very High

Example: 0x02 indicates very low battery and a Low Battery Alarm is sending.

5.4.1.6. GSM Signal Strength Levels

- 0x00: no signal;
- 0x01: extremely weak signal;
- 0x02: very weak signal;
- 0x03: good signal;
- 0x04: strong signal.

Example: 0x03 indicates the GSM signal is good.

5.4.1.7. Alarm/Language

0x00 (former bit) 0x01 (latter bit)

former bit: terminal alarm status (suitable for alarm packet and electronic fence project)

latter bit: the current language of the terminal

former bit	0x00: normal
	0x01: SOS
	0x02: Power Cut Alarm
	0x03: Shock Alarm
	0x04: Fence In Alarm
	0x05: Fence Out Alarm
latter bit	0x01: Chinese
	0x02: English

Examples:

No Alarm and Language is Chinese: 0x00 0x01

No Alarm and Language is English: 0x00 0x02

5.4.1.8. Information Serial Number

For details see Data Packet Format section 4.5.

5.4.1.9. Error Check

For details see Data Packet Format section 4.6.

5.4.1.10. Stop Bit

For details see Data Packet Format section 4.7.

5.4.2. Server Responds the Data Packet

	Description	Bits	Example
Login Message Packet (18 Byte)	Start Bit	2	<u>0x78 0x78</u>
	Packet Length	1	<u>0x05</u>
	Protocol Number	1	<u>0x01</u>
	Information Serial Number	2	<u>0x00 0x01</u>
	Error Check	2	<u>0xD9 0xDC</u>
	Stop Bit	2	<u>0x0D 0x0A</u>

The response packet from the server to the terminal: the protocol number in the response packet is identical to the protocol number in the data packet sent by the terminal.

5.4.2.1. Start Bit

For details see Data Packet Format section 4.1.

5.4.2.2. Packet Length

For details see Data Packet Format section 4.2.

5.4.2.3. Protocol Number

For details see Data Packet Format section 4.3.

5.4.2.4. Information Serial Number

For details see Data Packet Format section 4.5.

5.4.2.5. Error Check

For details see Data Packet Format section 4.6.

5.4.2.6. Stop Bit

For details see Data Packet Format section 4.7.

5.4.3. Examples

Example of data packet sent by the terminal							
78 78 0A 13 4B 04 03 00 01 00 11 06 1F 0D 0A							
Explain							
<u>0x78 0x78</u>	<u>0x0A</u>	<u>0x13</u>	<u>0x4B 0x04 0x03</u>	<u>0x00 0x01</u>	<u>0x00 0x11</u>	<u>0x06 0x1F</u>	<u>0x0D 0x0A</u>
Start Bit	Length	Protocol No.	Information Content	Reserved bit (Language)	Serial No.	Error Check	Stop Bit

Example of response packet returned by the server						
78 78 05 13 00 11 F9 70 0D 0A						
Explain						
<u>0x78 0x78</u>	<u>0x05</u>	<u>0x13</u>	<u>0x00 0x11</u>	<u>0xF9 0x70</u>	<u>0x0D 0x0A</u>	
Start Bit	Length	Protocol No.	Serial No.	Error Check	Stop Bit	

5.5. Calibration time Packet

a) terminel send request

Description	len	
Start bit	2	0x78 0x78
length	1	0x05
Protocol number	1	0x8A
Serial number	2	
Error check	2	
Stop bit	2	0x0D 0x0A

example: 78 78 05 8A 00 06 88 29 0D 0A

b) server reply

Description		len	
Start bit		2	0x78 0x78
length		1	0x0B
Protocol number		1	0x8A (UTC)
Message content	Date & time	6	year(1byte)month(1byte)day(1byte)hour(1byte)minute(1byte)second(1byte)
Serial number		2	
Error check		2	
Stop bit		2	0x0D 0x0A

example: 78 78 0B 8A 0F 0C 1D 00 00 15 00 06 F0 86 0D 0A

5.6. LBS Multi-Base Station Extended Information Packet

Description		len	详解
Start bit		2	0x78 0x78
length		1	0x3B
Protocol number		1	0x28
Message content	Date & time (UTC)	6	year (1byte) month (1byte) day (1byte) hour (1byte) minute (1byte) second (1byte)
	MCC	2	
	MNC	1	
	LAC	2	
	CI	3	
	RSSI	1	
	NLAC1	2	
	NCI1	3	
	NRSSI1	1	
	NLAC2	2	
	NCI2	3	
	NRSSI2	1	
	NLAC3	2	
	NCI3	3	
	NRSSI3	1	
	NLAC4	2	
	NCI4	3	
	NRSSI4	1	
	NLAC5	2	
	NCI5	3	
	NRSSI5	1	
	NLAC6	2	
	NCI6	3	
	NRSSI6	1	
	Time advance	1	
language	2	0x02	
Serial number		2	
Error check		2	
Stop bit		2	为 0x0D 0x0A

[illegible]

5.7. General Information Package

Description		len	
Start bit		2	0x79 0x79
length		2	0x20
Protocol number		1	0x94
Message content	Data type	1	0x00 external voltage 0x0A SIM card information 0xFD Voltage of analog input
	Data content	N	Refer to table below
Serial number		2	
Error check		2	
Stop bit		2	0x0D 0x0A

external voltage

When the type is 00, this bit transmits the external voltage, which is a two digit hexadecimal number. When the hexadecimal number is converted to decimal, divide by 100. For example: 0x04, 0x9f, **049f** is converted to decimal 1183, divide by 100 and it is 11.83, which means the external voltage value of the terminal is 11.83v.
device will send this packet every 10 minutes,and send it when connect to server,and when main power status change.

Example: 7979 0008 94 **00 049F** 0003 2D9F 0D0A

SIM card information

IMEI	8	If IMEI is 123456789123456, the IMEI section is 0x01 0x23 0x45 0x67 0x89 0x12 0x34 0x56
IMSI	8	If IMSI is 123456789123456, the IMIS section is 0x01 0x23 0x45 0x67 0x89 0x12 0x34 0x56
ICCID	10	If ICCID is 12345123456789123456, the ICCID section is 0X12 0x34 0x51 0x23 0x45 0x67 0x89 0x12 0x34 0x56

Example:

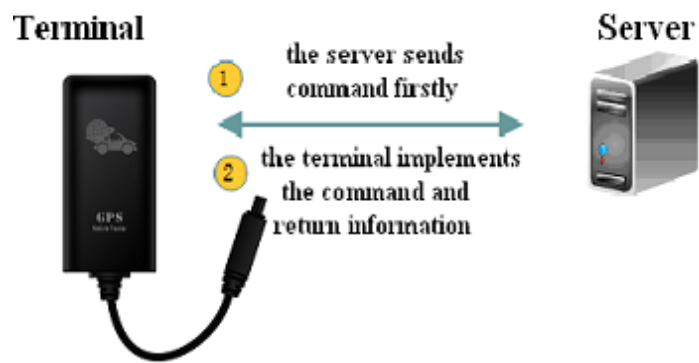
7979 0020 94 0A 0861702010034048 0460040515602540 898602B52616C0042540 0003 2D9F 0D0A

Voltage of analog input

When the type is FD, this bit transmits the voltage of analog input, which is a two digit hexadecimal number. When the hexadecimal number is converted to decimal, divide by 100. For example: 0x09, 0x7a, **097a** is converted to decimal 2426, divide by 100 and it is 24.26, which means the voltage value of analog input is 24.26v.
device will send this packet every 30 seconds

Example: 7979090094 **FD 097A** 000F692A0D0A

vi. Data Packet Sent From Server to Terminal



6.1. Packet Sent by Server

Format		Length (Byte)
Start Bit		2
Packet length		1
Protocol Number		1
Information Content	Length of Command	1
	Server Flag Bit	4
	Command Content	M
	Language	2
Information Serial Number		2
Error Check		2
Stop Bit		2

6.1.1. Start Bit

For details see Data Packet Format section 4.1.

6.1.2. Packet Length

For details see Data Packet Format section 4.2.

6.1.3. Protocol Number

The Protocol Number of terminal transmission is 0x80.

6.1.4. Length of Command

Server Flag Bit + Length of Command Content

Example: measured in bytes, 0x0A means the content of command occupied ten bytes.

6.1.5. Server Flag Bit

It is reserved to the identification of the server. The binary data received by the terminal is returned without change.

6.1.6. Command Content

It is represented in ASC II of string, and the command content is compatible with SMS command. Such as RELAY,1#

6.1.7. Language

A bit indicates the current language used in the terminal.

Chinese: 0x00 0x01

English: 0x00 0x02

6.1.8. Information Serial Number

For details see Data Packet Format section 4.5.

6.1.9. Error Check

For details see Data Packet Format section 4.6.

6.1.10. Stop Bit

For details see Data Packet Format section 4.7.

Example:

SMS command: cut off fuel supply

RELAY,1#

SMS reply:

Cut off the fuel supply:Success!

GPRS command: cut off fuel supply

0x787814800C0000000052454C41592C31230001000327960D0A

GPRS reply:

0x78782C152400000000437574206F666620746865206675656C20737570706C793A5375636365737321000200091D9E0D0A

SMS command: resume

RELAY,0#

SMS reply:

cancelled the command of cutting-off fuel supply; fuel supply is on.

GPRS command: resume

0x787814800C0000000052454C41592C30230001000457020D0A

GPRS reply:

0x78785015480000000063616E63656C6C65642074686520636F6D6D616E64206F662063757474696E672D6F6666206675656C20737570706C793B206675656C20737570706C79206973206F6E2E0002000CF5D90D0A

6.2. Packet Replied by Terminal

Format		Length (Byte)
Start Bit		2
Packet Length		1
Protocol Number		1
Information Content	Length of Command	1
	Server Flag Bit	4
	Command Content	M
	Language	2
Information Serial Number		2
Error Check		2
Stop Bit		2

6.2.1. Start Bit

For details see Data Packet Format section 4.1.

6.2.2. Packet Length

For details see Data Packet Format section 4.2.

6.2.3. Protocol Number

The terminal responds to the command sent by the server. The format of data packet is consistent with “the command sent by the server to the terminal”, but the Protocol Number herein is different and is 0x15.

6.2.4. Length of Command

Server Flag Bit + Length of Command Content

Example: measured in bytes, 0x0A means the content of command occupied ten bytes.

6.2.5. Server Flag Bit

It is reserved to the identification of the server. The binary data received by the terminal is returned without change.

6.2.6. Command Content

It is represented in ASC II of string, and the command content is compatible with Scout text message command.

6.2.7. Language

A bit indicates the current language used in the terminal.

Chinese: 0x00 0x01

English: 0x00 0x02

6.2.8. Information Serial Number

For details see Data Packet Format section 4.5.

6.2.9. Error Check

For details see Data Packet Format section 4.6.

6.2.10. Stop Bit

For details see Data Packet Format section 4.7.

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vii. Appendix A: code fragment of the CRC-ITU lookup table algorithm implemented based on C language

Code fragment of the CRC-ITU lookup table algorithm implemented based on C language is as follow:

```
static const U16 crctab16[] =
{
    0X0000, 0X1189, 0X2312, 0X329B, 0X4624, 0X57AD, 0X6536, 0X74BF,
    0X8C48, 0X9DC1, 0XAF5A, 0XBED3, 0XCA6C, 0XDBE5, 0XE97E, 0XF8F7,
    0X1081, 0X0108, 0X3393, 0X221A, 0X56A5, 0X472C, 0X75B7, 0X643E,
    0X9CC9, 0X8D40, 0XBFDB, 0XAE52, 0XDAED, 0XCB64, 0XF9FF, 0XE876,
    0X2102, 0X308B, 0X0210, 0X1399, 0X6726, 0X76AF, 0X4434, 0X55BD,
    0XAD4A, 0XBCC3, 0X8E58, 0X9FD1, 0XEB6E, 0XFAE7, 0XC87C, 0XD9F5,
    0X3183, 0X200A, 0X1291, 0X0318, 0X77A7, 0X662E, 0X54B5, 0X453C,
    0XBDCB, 0XAC42, 0X9ED9, 0X8F50, 0XFBF7, 0XEA66, 0XD8FD, 0XC974,
    0X4204, 0X538D, 0X6116, 0X709F, 0X0420, 0X15A9, 0X2732, 0X36BB,
    0XCE4C, 0XD5C5, 0XED5E, 0XFCDD, 0X8868, 0X99E1, 0XAB7A, 0XBAF3,
    0X5285, 0X430C, 0X7197, 0X601E, 0X14A1, 0X0528, 0X37B3, 0X263A,
    0XDECD, 0XCF44, 0XFFDD, 0XEC56, 0X98E9, 0X8960, 0XBBFB, 0XAA72,
    0X6306, 0X728F, 0X4014, 0X519D, 0X2522, 0X34AB, 0X0630, 0X17B9,
    0XEF4E, 0XFEC7, 0XCC5C, 0XDDD5, 0XA96A, 0XB8E3, 0X8A78, 0X9BF1,
    0X7387, 0X620E, 0X5095, 0X411C, 0X35A3, 0X242A, 0X16B1, 0X0738,
    0XFFCF, 0XEE46, 0XDCDD, 0XCD54, 0XB9EB, 0XA862, 0X9AF9, 0X8B70,
    0X8408, 0X9581, 0XA71A, 0XB693, 0XC22C, 0XD3A5, 0XE13E, 0XF0B7,
    0X0840, 0X19C9, 0X2B52, 0X3ADB, 0X4E64, 0X5FED, 0X6D76, 0X7CFF,
    0X9489, 0X8500, 0XB79B, 0XA612, 0XD2AD, 0XC324, 0XF1BF, 0XE036,
    0X18C1, 0X0948, 0X3BD3, 0X2A5A, 0X5EE5, 0X4F6C, 0X7DF7, 0X6C7E,
    0XA50A, 0XB483, 0X8618, 0X9791, 0XE32E, 0XF2A7, 0XC03C, 0XD1B5,
    0X2942, 0X38CB, 0X0A50, 0X1BD9, 0X6F66, 0X7EEF, 0X4C74, 0X5DFD,
    0XB58B, 0XA402, 0X9699, 0X8710, 0XF3AF, 0XE226, 0XD0BD, 0XC134,
    0X39C3, 0X284A, 0X1AD1, 0X0B58, 0X7FE7, 0X6E6E, 0X5CF5, 0X4D7C,
    0XC60C, 0XD785, 0XE51E, 0XF497, 0X8028, 0X91A1, 0XA33A, 0XB2B3,
    0X4A44, 0X5BCD, 0X6956, 0X78DF, 0X0C60, 0X1DE9, 0X2F72, 0X3EFB,
    0XD68D, 0XC704, 0XF59F, 0XE416, 0X90A9, 0X8120, 0XB3BB, 0XA232,
    0X5AC5, 0X4B4C, 0X79D7, 0X685E, 0X1CE1, 0X0D68, 0X3FF3, 0X2E7A,
    0XE70E, 0XF687, 0XC41C, 0XD595, 0XA12A, 0XB0A3, 0X8238, 0X93B1,
    0X6B46, 0X7ACF, 0X4854, 0X59DD, 0X2D62, 0X3CEB, 0X0E70, 0X1FF9,
    0XF78F, 0XE606, 0XD49D, 0XC514, 0XB1AB, 0XA022, 0X92B9, 0X8330,
    0X7BC7, 0X6A4E, 0X58D5, 0X495C, 0X3DE3, 0X2C6A, 0X1EF1, 0X0F78,
};

// calculate the 16-bit CRC of data with predetermined length.
U16 GetCrc16(const U8* pData, int nLength)
{
    U16 fcs = 0xffff;           // initialization
    while(nLength>0){
        fcs = (fcs >> 8) ^ crctab16[(fcs ^ *pData) & 0xff];
        nLength--;
        pData++;
    }
    return ~fcs;               // negated
}
```